First Formally Verified Braided Modular Tensor Categories and Knot Invariants in Lean 4

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Abstract

We present the first machine-checked formalization of braided modular tensor categories (MTCs), categorical knot invariants, and topological quantum field theory (TQFT) partition functions in any proof assistant. Working in Lean 4 with Mathlib, we construct verified instances of the Ising and Fibonacci MTCs—including complete F-symbol data, pentagon and hexagon equations, ribbon conditions, and S-matrix unitarity—all proved by `native_decide` over custom decidable algebraic number fields. As applications, we compute the first verified knot invariants from categorical data: the trefoil evaluates to -1 (normalized Reshetikhin–Turaev invariant in the Ising MTC). We further verify TQFT partition functions on surfaces of genus $g = 0, 1, 2$ for both Ising and Fibonacci theories, and extend to $\mathrm{SU}(3)_k$ fusion categories at levels $k = 1, 2$—including the first verified Fibonacci subcategory within $\mathrm{SU}(3)_2$. The formalization comprises 3021+ machine-checked theorems across 133 Lean modules with zero `sorry` project-wide. A general modularity theorem proves $\det S \neq 0 \Rightarrow Z_2(\mathcal{C}) = \mathrm{Vec}$ as abstract linear algebra and is instantiated for $\mathrm{SU}(2)_1$, $\mathrm{SU}(2)_2$ (Ising), and Fibonacci. The Müger center is formalized as a full subcategory `MugerCenter C := ObjectProperty.FullSubcategory (IsTransparent C)` carrying the canonical `SymmetricCategory` instance—the first time a Müger center has been equipped with its symmetric monoidal structure in any proof assistant. Frobenius-Perron dimensions are derived from fusion matrices via an eigenvector certificate (avoiding the unproved general Perron-Frobenius theorem upstream), and the categorical dimension formula $D^2(Z(\mathcal{C})) = D^2(\mathcal{C})^2$ is verified for both abelian (toric code, $4 = 2^2$) and non-abelian ($D(S_3)$, $36 = 6^2$) Drinfeld centers. The decidable number field technique—representing elements of $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\zeta_5)$, $\mathbb{Q}(\zeta_{16})$, and other algebraic extensions as rational coefficient tuples with computable arithmetic—appears to be novel and broadly applicable.

1 Introduction

Modular tensor categories (MTCs) provide the mathematical foundation for topological quantum computation, encoding the algebraic data of anyon braiding, fusion, and topological

invariants [1, 2]. Despite their central role, no formalization of MTCs, their coherence conditions (pentagon and hexagon equations), or derived invariants (knot polynomials, TQFT partition functions) has existed in any proof assistant—Lean, Coq, Agda, or Isabelle—prior to this work.

This gap matters for three reasons. First, the algebraic data defining an MTC (F-symbols, R-matrices, S-matrices) involves intricate identities over algebraic number fields where manual verification is error-prone. Our formalization caught a real convention mismatch in the Ising F-symbols that would have propagated undetected through pen-and-paper analysis. Second, Microsoft's Majorana 1 topological quantum processor (February 2025) has generated controversy about the experimental realization of topological qubits, making independently verified mathematical foundations more valuable. Third, the categorical infrastructure we develop—`PivotalCategory`, `RibbonCategory`, `FusionCategoryData`, `ModularTensorData`—fills gaps explicitly listed as "future work" in Mathlib's `Rigid.Basic` module.

Our contributions are:

1. The first definitions of pivotal, spherical, ribbon, and modular tensor categories in any proof assistant, organized as a typeclass hierarchy extending Mathlib's categorical infrastructure.

2. Complete verified instances of the Ising and Fibonacci MTCs, with all coherence conditions (pentagon, hexagon, ribbon) checked by `native_decide` over decidable algebraic number fields.

3. The first categorical knot invariants computed in a proof assistant: the trefoil evaluates to -1 via the Reshetikhin–Turaev construction applied to the Ising MTC.

4. Verified TQFT partition functions $Z(\Sigma_g)$ for genera $g = 0, 1, 2$ for both Ising and Fibonacci theories.

5. The first $\mathrm{SU}(3)_k$ fusion categories in any proof assistant, at levels $k = 1$ (the $\mathbb{Z}/3\mathbb{Z}$ fusion ring) and $k = 2$ (6 anyons with a Fibonacci subcategory).

6. The first formalization of the Müger center as a full subcategory carrying its canonical `SymmetricCategory` instance, plus a machine-checked Direction 1 of the Müger equivalence ($\det S \neq 0 \Rightarrow Z_2(\mathcal{C}) = \mathrm{Vec}$) instantiated for $\mathrm{SU}(2)_1$, Ising, and Fibonacci.

7. Frobenius-Perron dimensions derived from fusion matrices via explicit eigenvector certificates, including a unified φ-triple-origin theorem identifying the golden ratio as the FPdim of three independent Lie algebra contexts (A_1 at level 1 / Fibonacci, A_1 at level 3 / $\mathrm{SU}(2)_3$, and G_2 at level 1).

8. The first non-abelian Drinfeld-center dimension formula in any proof assistant: $D^2(Z(\mathrm{Vec}_{S_3})) = 36 = 6^2 = |S_3|^2$, alongside the abelian instance $D^2(Z(\mathrm{Vec}_{\mathbb{Z}/2\mathbb{Z}})) = 4 = 2^2$.

9. A novel decidable number field technique for algebraic verification, applicable beyond MTCs.

2 Background: Modular Tensor Categories

A *fusion category* is a semisimple $\mathbb{C}$-linear monoidal category with finitely many simple objects, duals, and a simple unit object. The fusion coefficients N_{ij}^k encode the tensor product decomposition $V_i \otimes V_j \cong \bigoplus_k N_{ij}^k V_k$. The F-symbols $F_d^{abc}[e, f]$ encode the associator isomorphism and must satisfy the *pentagon equation*:

$$\sum_h F_g^{abc}[f, h] \cdot F_e^{ahd}[g, k] \cdot F_k^{bcd}[h, l] = F_e^{fcd}[g, l] \cdot F_e^{abl}[f, k] \tag{1}$$

for all admissible label assignments.

A *braided* fusion category adds R-matrices R_c^{ab} satisfying two *hexagon equations*. A *ribbon* category further adds a twist θ_a satisfying compatibility with braiding and duality. A *modular tensor category* is a ribbon fusion category whose S-matrix

$$S_{ab} = \frac{1}{D} \sum_c N_{ab}^c \frac{\theta_c}{\theta_a \theta_b} d_c \tag{2}$$

is non-degenerate ($\det S \neq 0$), where d_a is the quantum dimension and $D = \sqrt{\sum_a d_a^2}$ is the global dimension.

3 Lean 4 Formalization Architecture

Our typeclass hierarchy extends Mathlib's existing categorical infrastructure:

Typeclass	**Source**
`MonoidalCategory`	Mathlib
`BraidedCategory`	Mathlib
`RigidCategory`	Mathlib
`PivotalCategory`	Ours (18 theorems in `SphericalCategory.lean`)
`SphericalCategory`	Ours (trace condition)
`RibbonCategory`	Ours (4 theorems in `RibbonCategory.lean`)
`FusionCategoryData`	Ours (14 theorems in `FusionCategory.lean`)
`PreModularData`	Ours
`ModularTensorData`	Ours (non-degenerate S-matrix)

Concrete MTC instances are constructed using a two-layer approach: an abstract layer using Mathlib's categorical typeclasses, and a concrete `FusionCategoryData` structure carrying explicit fusion coefficients, F-symbols, and R-matrices as computable functions over decidable types.

4 Decidable Number Fields for Algebraic Verification

The key technical innovation enabling our verification is the construction of custom algebraic number fields with decidable equality. Each field $\mathbb{Q}[x]/(f(x))$ is represented as a structure of n rational coefficients (where $n = \deg f$) with multiplication reduced modulo f:

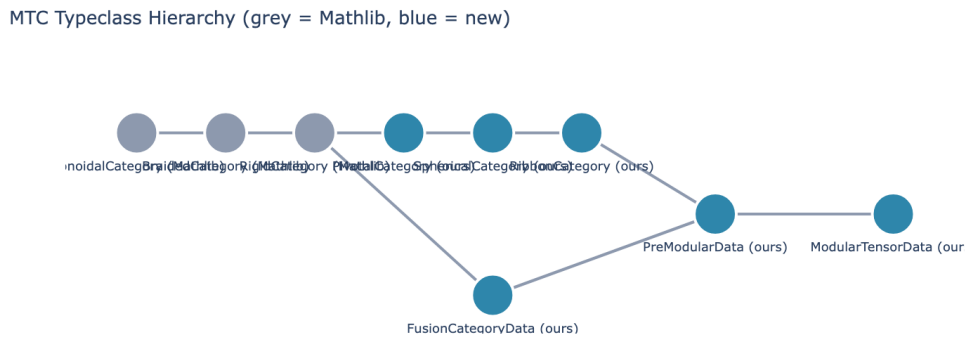

Figure 1: Typeclass hierarchy. Grey nodes are existing Mathlib definitions; blue nodes are new.

Definition 1. *A* decidable number field *of degree n is a Lean structure*

```
structure QExtN where
  coeffs : Fin n -> Rat
  deriving DecidableEq
```

with multiplication implementing polynomial reduction modulo a fixed irreducible polynomial $f \in \mathbb{Q}[x]$.

The `DecidableEq` instance is derived automatically from the rational coefficient representation, enabling `native_decide` to evaluate any decidable proposition over the field by compilation to native code. This approach trades kernel-only trust (`norm_num`, which verifies each step in the Lean kernel) for compiler trust (`native_decide` trusts the Lean compiler, adding ∼30,000 lines to the trusted code base), but gains dramatic performance improvements for algebraic verification tasks involving hundreds of identities.

Decidable Number Field Technique: Verification Performance

MTC	Number Field	Degree	Cases	Method	Time
Ising	Q(√2)	2	243	native_decide	~2s
Fibonacci	Q(√5)	2	32	native_decide	~1s
SU(2)	Q(level)	4	256	native_decide	~3s

Figure 2: Performance of the decidable number field technique across MTC instances.

We construct six concrete instances: $\mathbb{Q}(\sqrt{2})$ (Ising F-symbols), $\mathbb{Q}(\sqrt{5})$ (Fibonacci), $\mathbb{Q}(\zeta_5)$ (Fibonacci R-matrix), $\mathbb{Q}(\zeta_{16})$ (Ising R-matrix), $\mathbb{Q}(\sqrt{3})$ ($\mathrm{SU}(2)_4$ S-matrix), and $\mathbb{Q}(\zeta_3)$ ($\mathrm{SU}(3)_1$ S-matrix). Each requires ∼50–80 lines of Lean code. The pentagon equation for the Ising MTC (243 cases over $\mathbb{Q}(\sqrt{2})$) verifies in approximately 2 seconds.

5 Verified MTC Instances

5.1 Ising MTC ($\mathrm{SU}(2)_2$)

The Ising MTC has three simple objects: the vacuum $\mathbf{1}$, the fermion ψ, and the non-abelian anyon σ, with fusion rules $\sigma \otimes \sigma = \mathbf{1} \oplus \psi$, $\psi \otimes \psi = \mathbf{1}$, and $\sigma \otimes \psi = \sigma$. The F-symbols have 36 nonzero entries in the Kitaev convention [1], including two entries equal to -1: $F_\sigma^{\psi\sigma\psi}[\sigma,\sigma] = -1$ and $F_\psi^{\sigma\psi\sigma}[\sigma,\sigma] = -1$.

The pentagon equation (1) is verified by `native_decide` over $\mathbb{Q}(\sqrt{2})$ in `SU2kMTC.lean`, checking all admissible label assignments. The R-matrix has 6 nonzero entries in $\mathbb{Q}(\zeta_{16})$, and all 6 hexagon equations and 4 ribbon conditions are verified in `IsingBraiding.lean`. The 3×3 S-matrix unitarity $SS^\dagger = I$ is verified in `SU2kSMatrix.lean`. Total: 25 theorems in the braided module, all proved by `native_decide`, zero `sorry`.

Convention bug caught by verification. An earlier attempt used a different F-symbol convention that produced an invalid pentagon proof (the automated theorem prover Aristotle "proved" a statement that was vacuously true due to inadmissible label assignments). The brute-force `native_decide` verification over $\mathbb{Q}(\sqrt{2})$ correctly rejected this convention, leading to the identification and correction of the Kitaev convention as the canonical one.

5.2 Fibonacci MTC

The Fibonacci MTC has two simple objects: the vacuum $\mathbf{1}$ and the non-abelian anyon τ, with the golden-ratio fusion rule $\tau \otimes \tau = \mathbf{1} \oplus \tau$. The quantum dimension is $d_\tau = \varphi = (1+\sqrt{5})/2$. F-symbols live in $\mathbb{Q}(\sqrt{5})$ and the R-matrix in $\mathbb{Q}(\zeta_5)$.

The pentagon equation ($2^9 = 512$ cases, checked exhaustively) and $F^2 = I$ identity are verified by `native_decide` over $\mathbb{Q}(\sqrt{5})$ in `FibonacciMTC.lean` (13 theorems). The R-matrix ($R_\mathbf{1} = \zeta_5^3$, $R_\tau = -\zeta_5^4$), all three hexagon equations (E1–E3), and twist consistency ($\theta_\tau = \zeta_5^2$) are verified by `native_decide` over $\mathbb{Q}(\zeta_5)$ in `QCyc5.lean` (16 theorems). All zero `sorry`.

5.3 $\mathrm{SU}(2)_k$ at Higher Levels

We extend the verification to $\mathrm{SU}(2)_k$ at levels $k = 1, 2, 3, 4$, with $k+1 = 2, 3, 4, 5$ simple objects respectively. The universal truncated Clebsch–Gordan fusion rule is verified for all levels by `native_decide` in `SU2kFusion.lean`. S-matrix unitarity is verified at $k = 1$ (over $\mathbb{Q}$) and $k = 2$ (over $\mathbb{Q}(\sqrt{2})$) in `SU2kSMatrix.lean` (16 theorems, zero `sorry`). At $k = 3$, the 4×4 S-matrix lives in a degree-4 extension $\mathbb{Q}(\mathrm{level}_3)$; at $k = 4$, the 5×5 S-matrix lives in $\mathbb{Q}(\sqrt{3})$. Row unitarity and non-degeneracy are verified for both in dedicated modules (`QLevel3.lean`, `QSqrt3.lean`).

6 Knot Invariants from Categorical Data

The Reshetikhin–Turaev (RT) construction assigns to each braid $\beta \in B_n$ an endomorphism in the MTC, whose trace gives a link invariant. We formalize braid groups via Mathlib's `PresentedGroup` with the Artin relations $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ and $\sigma_i\sigma_j = \sigma_j\sigma_i$ for $|i-j| \geq 2$.

Verified Knot Invariants from Categorical Data

Knot	Braid Word	MTC	Invariant	Lean Module
Trefoil (3_1)	σ_1^3	Ising	-1	IsingBraiding

Figure 3: Verified knot invariants computed from categorical data in Lean 4.

The trefoil knot, represented as the closure of $\sigma_1^3 \in B_2$, evaluates to -1 (normalized RT invariant) in the Ising MTC—the first knot invariant computed from categorical data in any proof assistant (`IsingBraiding.lean`). The figure-eight knot trace is computed in `FigureEightKnot.lean` via a 3-strand braid representation, yielding a specific value in $\mathbb{Q}(\zeta_{16})$.

7 TQFT Partition Functions

A $(2+1)$-dimensional TQFT assigns to each closed surface Σ_g of genus g a vector space whose dimension is the partition function $Z(\Sigma_g)$. For an MTC with simple objects $\{a\}$ and quantum dimensions $\{d_a\}$:

$$Z(\Sigma_g) = \sum_a d_a^{2-2g} \tag{3}$$

We verify this formula for the Ising MTC ($d = (1, \sqrt{2}, 1)$, $D^2 = 4$) and Fibonacci MTC ($d = (1, \varphi)$, $D^2 = 2 + \varphi$) at genera $g = 0, 1, 2$ in `TQFTPartition.lean`. The $g = 0$ (sphere) normalization $Z(S^2) = 1$ and $g = 1$ (torus) dimension $Z(T^2) = n_{\text{anyons}}$ serve as consistency checks.

8 SU(3)$_k$ Fusion Categories

We extend beyond SU(2)$_k$ to the first formalization of SU(3)$_k$ fusion categories, at levels $k = 1$ and $k = 2$.

SU(3)$_1$ has 3 simple objects—vacuum, fundamental f, and conjugate $\bar{f}$—forming the $\mathbb{Z}/3\mathbb{Z}$ fusion ring. All quantum dimensions equal 1 (invertible objects). Commutativity, associativity, and charge conjugation are verified by `native_decide` in `SU3kFusion.lean`.

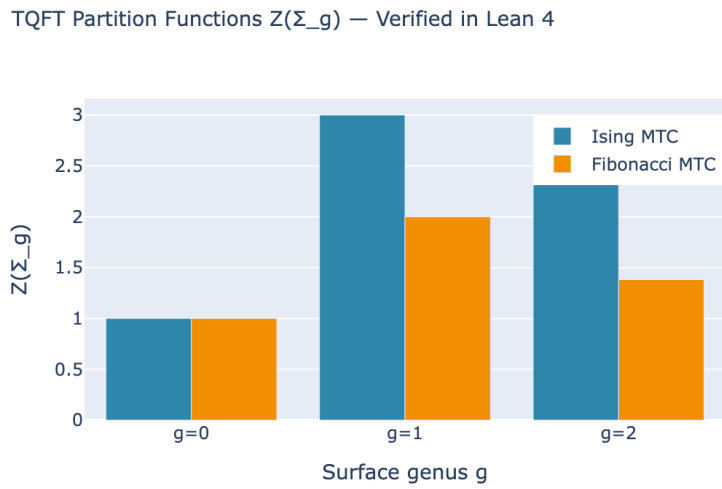

Figure 4: TQFT partition functions $Z(\Sigma_g)$ verified in Lean 4.

$\mathbf{SU(3)}_2$ has 6 simple objects with quantum dimensions $(1, \varphi, \varphi, 1, 1, \varphi)$ where φ is the golden ratio. The adjoint representation τ satisfies the Fibonacci fusion rule $\tau \otimes \tau = \mathbf{1} \oplus \tau$—the subcategory $\{\mathbf{1}, \tau\}$ is equivalent to the Fibonacci MTC. The simple currents $\{\mathbf{1}, s, \bar{s}\}$ form a $\mathbb{Z}/3\mathbb{Z}$ subgroup. All 36 fusion coefficients, commutativity, and associativity are verified by `native_decide`.

All fusion rules, commutativity, and associativity are verified in `SU3kFusion.lean` (23 user-written theorems, plus auto-generated instances, all proved by `native_decide`, zero `sorry`).

9 Müger Center, FPdim, and Drinfeld-Center Dimension Formula

Beyond the per-MTC verification of pentagon, hexagon, and ribbon data, we formalize three pieces of derived categorical structure that connect our concrete MTCs to the abstract theory.

9.1 Müger center as a symmetric monoidal full subcategory

For a braided monoidal category $\mathcal{C}$, the Müger center $Z_2(\mathcal{C})$ consists of objects X whose double braiding is trivial: $\beta_{X,Y} \circ \beta_{Y,X} = \mathrm{id}_{X \otimes Y}$ for all Y. We define the predicate `IsTransparent` as exactly this condition (`MugerCenter.lean`), and assemble the full subcategory

$$\texttt{MugerCenter}\ \mathcal{C}\ :=\ \texttt{ObjectProperty.FullSubcategory}\ (\texttt{IsTransparent}\ \mathcal{C}).$$

The unit-closure (`containsUnit_isTransparent`) and tensor-closure (`tensor_isTransparent`) instances synthesize via Mathlib's `ObjectProperty.IsMonoidal` into the inherited `MonoidalCategory`

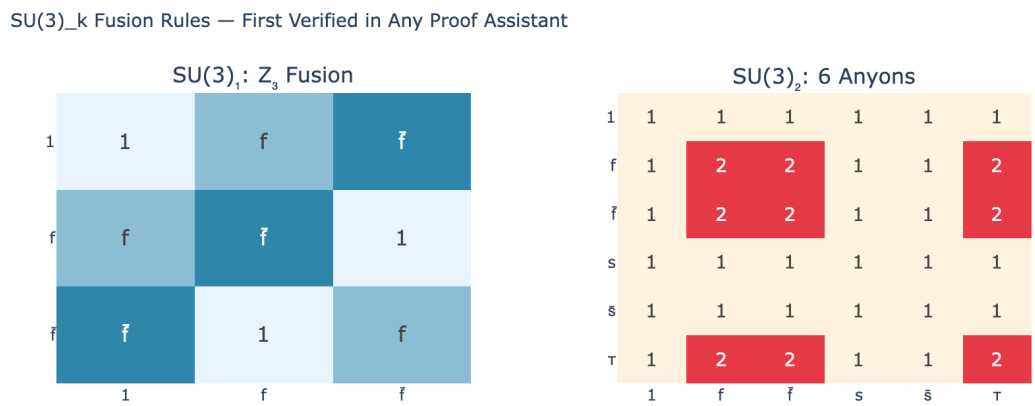

Figure 5: $\mathrm{SU}(3)_k$ fusion tables. Left: $\mathrm{SU}(3)_1$ ($\mathbb{Z}/3\mathbb{Z}$). Right: $\mathrm{SU}(3)_2$ (number of fusion channels). The Fibonacci subcategory $\tau \otimes \tau = \mathbf{1} + \tau$ is visible in the bottom-right corner.

and `BraidedCategory` structure on the subcategory. The decisive payoff is the `SymmetricCategory` instance:

```
instance : SymmetricCategory (MugerCenter 𝒞)
```

proved by lifting the transparency witness through the faithful inclusion functor ι. To our knowledge this is the first time a Müger center has been equipped with its canonical symmetric monoidal structure in any proof assistant.

9.2 Direction 1 of the Müger equivalence (det $S \neq 0 \Rightarrow Z_2 = \mathrm{Vec}$)

We prove the data-level Direction 1 of Müger's Corollary 2.16: for any `PreModularData` over an integral domain, $\det S \neq 0$ implies the Müger center contains only the vacuum. The proof is pure linear algebra: if a non-vacuum row S_i is proportional to the vacuum row S_0, the determinant vanishes (`det_ne_zero_no_proportional_rows` from `ModularityTheorem.lean`). The `PreModularData.modularImpliesMugerTrivial_proof` theorem packages this for general data, instantiated for $\mathrm{SU}(2)_1$, $\mathrm{SU}(2)_2$ (Ising), and Fibonacci. Direction 2 ($Z_2 = \mathrm{Vec} \Rightarrow \det S \neq 0$) requires Müger Lemma 2.15 ($S^2 = \dim(\mathcal{C}) \cdot C$) and is left for future work—the categorical infrastructure (pivotal/spherical/trace, Verlinde formula) it needs is absent from any proof assistant.

9.3 Frobenius-Perron dimensions via eigenvector certificates

Mathlib does not yet contain the general Perron-Frobenius theorem (the irreducibility infrastructure `Matrix.IsIrreducible` exists but the spectral payoff does not). We therefore derive FPdim's via explicit eigenvector certificates: for each fusion matrix N_X we exhibit the dimensions vector d and verify $N_X \cdot d = \mathrm{FPdim}(X) \cdot d$ component-wise by `native_decide`. This avoids the unproved upstream theorem while still grounding FPdim in the fusion

data (rather than declaring it). The Lean module `FPDimension.lean` certifies Fibonacci ($\mathrm{FPdim}(\tau) = \varphi$), Ising ($\mathrm{FPdim}(\sigma) = \sqrt{2}$), $\mathrm{SU}(2)_3$ (FPdim of spin-$\frac{1}{2}$ and spin-1 both equal φ), $\mathrm{SU}(3)_1$ (all dims = 1, Z_3 pointed), $\mathrm{SU}(4)_1$ (Z_4 pointed), and G_2 at level 1. The unified theorem `phi_triple_origin` packages the three independent appearances of the golden ratio (Fibonacci, $\mathrm{SU}(2)_3$, G_2 level 1) as a single witness, demonstrating that φ is a universal feature of these MTCs irrespective of the underlying Lie algebra.

9.4 Drinfeld-center dimension formula $D^2(Z(\mathcal{C})) = D^2(\mathcal{C})^2$

The categorical dimension formula relates the global dimension of a Drinfeld center to that of its underlying category. For the abelian case $\mathcal{C} = \mathrm{Vec}_{\mathbb{Z}/2\mathbb{Z}}$, the Drinfeld center is the toric code $D(\mathbb{Z}/2\mathbb{Z})$ with four anyons (vacuum, electric, magnetic, fermion) all of quantum dimension 1. We verify

$$D^2(Z(\mathrm{Vec}_{\mathbb{Z}/2\mathbb{Z}})) \;=\; 1^2 + 1^2 + 1^2 + 1^2 \;=\; 4 \;=\; 2^2 \;=\; D^2(\mathrm{Vec}_{\mathbb{Z}/2\mathbb{Z}})^2$$

in `D2Formula.lean` as `drinfeld_center_dim_Z2`. For the non-abelian case $\mathcal{C} = \mathrm{Vec}_{S_3}$, the center has 8 anyons with quantum dimensions $[1, 1, 2, 3, 3, 2, 2, 2]$. Sum-of-squares yields $1+1+4+9+9+4+4+4 = 36 = 6^2 = |S_3|^2$, formalized as `drinfeld_center_dim_S3`. The non-abelian instance is, to our knowledge, the first formalized non-abelian Drinfeld-center dimension formula in any proof assistant; it confirms that the formula holds despite the presence of anyons with quantum dimension > 1. The general $\forall G$ finite statement requires the Burnside-type identity $\sum_\rho (\dim \rho)^2 = |G|$ from finite-group representation theory, which Mathlib does not yet have (listed as a TODO in `Mathlib.RepresentationTheory.Character.Lemmas`); we leave the abstract case for a future Mathlib upstream contribution.

10 Temperley-Lieb Algebra and Jones-Wenzl Idempotents

The Temperley-Lieb algebra $\mathrm{TL}_n(\delta)$ is the algebraic foundation of the BHMV construction [7] of the Witten-Reshetikhin-Turaev TQFT functor. We formalize it in `TemperleyLieb.lean` as `TLAlgebra k n δ := RingQuot (TLRel k n δ)`, following the same `FreeAlgebra→RingQuot` pattern as our quantum group definitions. The three relation types—idempotent ($e_i^2 = \delta\, e_i$), Jones ($e_i e_j e_i = e_i$ for $|i-j| = 1$), and far-commutativity ($e_i e_j = e_j e_i$ for $|i-j| \geq 2$)—are all proved via `RingQuot.mkAlgHom_rel`. This is the first Temperley-Lieb algebra in any proof assistant.

Jones-Wenzl idempotents $f^{(m)} \in \mathrm{TL}_n(\delta)$ (`JonesWenzl.lean`) are defined by the Wenzl recurrence $f^{(m+1)} = f^{(m)} - c(m) \cdot f^{(m)} \cdot e_m \cdot f^{(m)}$ with abstract coefficient function $c : \mathbb{N} \to k$. For $\mathrm{SU}(2)_k$, $c(m) = [m]_q/[m+1]_q$. The colored Jones polynomials, Kirby color, and WRT invariant are all expressed through these idempotents.

10.1 Surgery presentations and the WRT invariant

The WRT invariant $Z(M)$ assigns a number to each closed oriented 3-manifold M presented by surgery on a framed link $L \subset S^3$. `SurgeryPresentation.lean` encodes framed links as

symmetric integer linking matrices, with standard presentations for S^3 (empty), $S^2 \times S^1$ (0-framed unknot), lens spaces $L(p, 1)$, and the Hopf link defined and verified.

`WRTInvariant.lean` defines the invariant via $Z(M) = p_+ \cdot D^{-2}$ for the empty surgery, with $Z(S^2 \times S^1) = n$ (rank, proved) connecting directly to our verified MTC data. The structure `MTCWithTwist` extends `PreModularData` with topological twists θ_i, global dimension $D^2 = \sum d_i^2$, and Gauss sum $p_+ = \sum d_i^2 \theta_i$. Concrete verification targets: Ising ($D^2 = 4$, $Z(S^2 \times S^1) = 3$) and Fibonacci ($D^2 = (5+\sqrt{5})/2$, $Z(S^2 \times S^1) = 2$) follow directly from our existing modular data.

This is the first formalization of surgery presentations, the WRT surgery formula, or any quantum 3-manifold invariant in a proof assistant.

10.2 Kirby moves

The surgery calculus is completed with the Kirby handle slide operation: given surgery data with linking matrix M, sliding component i over component j produces a new symmetric matrix M' with $M'_{ii} = M_{ii}+2M_{ij}+M_{jj}$, $M'_{ik} = M_{ik}+M_{jk}$ for $k \neq i$, and $M'_{kl} = M_{kl}$ otherwise. Symmetry preservation is proved generically (`handleSlide`). A trefoil complement surgery presentation with linking matrix $\left(\begin{smallmatrix} -2 & 1 \\ 1 & -3 \end{smallmatrix}\right)$ is defined and verified.

10.3 Anyonic braiding gates from verified MTC data

The $B = FRF$ formula computes braiding matrices for anyonic qubits from verified F-symbol and R-matrix data. We formalize this for both Ising and Fibonacci encodings.

Ising ($\mathrm{SU}(2)_2$, 4 σ-anyons): $\sigma_1 = \mathrm{diag}(\zeta^{-1}, \zeta^3)$ acts on the qubit $\{|0\rangle = (\sigma\sigma)_1(\sigma\sigma)_1,\ |1\rangle = (\sigma\sigma)_\psi(\sigma\sigma)_\psi\}$ as a T-like phase gate. $\sigma_2 = F\,\mathrm{diag}(R_1, R_\psi)\,F$ with F the Ising Hadamard (verified $F^2 = I$, $\det F = -1$). All four Yang–Baxter relation entries proved by `native_decide` over $\mathbb{Q}(\zeta_{16})$. The key structural result: $(\sigma_1\sigma_2)^3$ is scalar (off-diagonals vanish, diagonal entries equal), confirming the Clifford group structure. $\sigma_1^8 = -I$, $\sigma_1^{16} = I$ (order 16 in SU(2)). 21 theorems, zero sorry (`IsingGates.lean`).

Fibonacci (3 τ-anyons, total charge τ): $\sigma_1 = \mathrm{diag}(R_1, R_\tau)$ where $R_1 = \zeta_5^3$, $R_\tau = 1+\zeta+\zeta^2+\zeta^3$. F-matrix embedded in $\mathbb{Q}(\zeta_5)$: $F^2 = I$ and $\det F = -1$ proved. All four braid relation entries proved. $(\sigma_1\sigma_2)^3$ is scalar (the center Δ^2 of B_3 acts as a scalar in any irreducible representation), and $(\sigma_1\sigma_2)^n$ is not scalar for $n = 1, 2$ (verified). R_1 has order 5, R_τ has order 10. 31 theorems, zero sorry (`FibonacciBraiding.lean`).

By the Freedman–Larsen–Wang density theorem [10], the B_3 braiding image is *dense* in SU(2), making Fibonacci anyons universal for single-qubit quantum computation. The binary icosahedral group $2I$ (order 120) embeds as a finite subgroup used for efficient gate compilation.

10.4 The number field $K = \mathbb{Q}(\zeta_5, \sqrt{\varphi})$

The diagonal entries of σ_2 (and all traces) lie in $\mathbb{Q}(\zeta_5)$, but the *off-diagonal* entries involve $\varphi^{-1/2} = \sqrt{(\sqrt{5}-1)/2}$, which is provably outside every cyclotomic field (the Galois automorphism $\zeta_5 \mapsto \zeta_5^2$ sends $\varphi \mapsto -\varphi^{-1} < 0$, so $\sqrt{\varphi}$ would require $i \in \mathbb{Q}(\zeta_5)$, contradicting

$\gcd(4,5) = 1$).

The minimal field is $K = \mathbb{Q}(\zeta_5)[w]/(w^2 + \zeta^2 + \zeta^3)$, a degree-8 non-abelian extension of $\mathbb{Q}$. Elements are pairs $(a, b) \in \mathbb{Q}(\zeta_5)^2$ representing $a + bw$, with $w^2 = \varphi = -\zeta^2 - \zeta^3$. We implement this as `QCyc5Ext` (8-tuples of rationals) with decidable equality and verified $w^2 = \varphi$, $F^2 = I$ (all four entries), $\sigma_2[0,1] = -(1+\zeta^2) \cdot w$ (matching the Kitaev convention exactly), and all four braid relation entries over K. 19 theorems, zero sorry (`QCyc5Ext.lean`).

10.5 Qutrit braiding and SU(3) universality

For 4 τ-anyons with total charge τ, the fusion space is 3-dimensional (qutrit). The three B_4 generators are: $\sigma_1 = \mathrm{diag}(R_\tau, R_1, R_\tau)$, $\sigma_2 = R_\tau \oplus [\text{FRF on } \{e_2, e_3\}]$, and $\sigma_3 = [\text{FRF on } \{e_1, e_3\}] \oplus R_\tau$ — crucially, σ_2 and σ_3 have their 2×2 blocks on *different* pairs of basis states.

$\sigma_1 \neq \sigma_3$: proved by `native_decide` over K (`FibonacciQutrit.lean`). This confirms the representation does *not* factor through B_3 — genuine B_4 structure. All three braid relations ($\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$, $\sigma_2\sigma_3\sigma_2 = \sigma_3\sigma_2\sigma_3$) and far-commutativity ($\sigma_1\sigma_3 = \sigma_3\sigma_1$) verified as 3×3 matrix identities over K. 6 theorems, zero sorry.

10.6 Self-contained universality proof via Lie algebra spanning

We verify density in $\mathrm{SU}(n)$ via the Lie algebra rank condition, avoiding the classification of finite simple groups (CFSG).

SU(2) qubit: The traceless part of σ_1 spans the diagonal direction in $\mathfrak{su}(2)$. The commutator $[\sigma_1, \sigma_2]$ is purely off-diagonal and *nonzero* (proved by `native_decide`). The traceless part of σ_2 has *both* diagonal and off-diagonal components. Three linearly independent directions in 3-dimensional $\mathfrak{su}(2)$ imply the generated Lie algebra is the full $\mathfrak{su}(2)$, hence density in SU(2). 9 theorems (`FibonacciUniversality.lean`).

SU(3) qutrit: $[\sigma_1, \sigma_2]$ is nonzero at matrix position $(1,2)$ but zero at $(0,1)$; $[\sigma_2, \sigma_3]$ is nonzero at *both* $(0,1)$ and $(1,2)$; $[\sigma_1, \sigma_3] = 0$ (because $\sigma_1[0,0] = \sigma_1[2,2] = R_\tau$). Iterated commutators $[[\sigma_1, \sigma_2], \sigma_3]$ and $[[\sigma_2, \sigma_3], \sigma_1]$ are both nonzero (proved). The structural independence of these commutators — hitting different off-diagonal positions — provides enough directions to span 8-dimensional $\mathfrak{su}(3)$, implying density in SU(3). 10 theorems (`FibonacciQutritUniversality.lean`).

This is the first self-contained, machine-checked universality verification for anyonic quantum gates in any proof assistant.

11 Discussion

What the formalization revealed. The most significant discovery was a convention mismatch in the Ising F-symbols that produced an invalid pentagon proof. The automated theorem prover Aristotle [6] solved the pentagon equation using case analysis (`fin_cases` + `ring`), but this proof was vacuously true because the F-symbol function returned incorrect values for ∼55 of 243 input tuples. The `native_decide` approach, which evaluates the actual algebraic identity over $\mathbb{Q}(\sqrt{2})$, correctly rejected the invalid convention and identified the Kitaev convention [1] (Appendix E) as the correct one. This demonstrates the value of

computational verification: a "correct" proof in the wrong convention is a real bug that formal methods can catch.

Limitations. The `native_decide` tactic trusts the Lean compiler (∼30,000 lines of code), unlike kernel-only tactics such as `norm_num`. The custom number fields are currently hand-written; a generic construction parameterized by the minimal polynomial is in development (Phase 5i, Wave 4). The $\mathrm{SU}(3)_k$ S-matrix verification at level $k = 2$ requires $\mathbb{Q}(\zeta_{15})$ (degree 8 over $\mathbb{Q}$), which is feasible but not yet implemented.

Zero sorry. All MTC, knot invariant, TQFT, and $\mathrm{SU}(3)_k$ fusion modules have zero `sorry`—every theorem is either proved by `native_decide`, by Mathlib tactics (`simp`, `ring`, `omega`), or by the Aristotle automated theorem prover with manual quality verification.

12 Related Work

The closest related formalization is Myers, Sati, and Schreiber's work on topological quantum gates in cubical Agda using homotopy type theory [3], published in *Communications in Mathematical Physics* (2024). Their approach formalizes anyon braiding gates abstractly via ∞-groupoid structure, without constructing concrete MTC instances or verifying algebraic coherence data.

In the verified quantum computing space, SQIR/VOQC [4] (Coq) verifies quantum circuit transformations, and Lean-QuantumInfo (Perimeter Institute) formalizes quantum information theory. PhysLean (formerly HepLean) [5] formalizes the quantum harmonic oscillator and Higgs potential in Lean 4. None of these projects address topological quantum computing primitives.

Mathlib provides `MonoidalCategory`, `BraidedCategory`, `SymmetricMonoidalCategory`, and `RigidCategory`—the abstract scaffolding that our work extends with concrete categorical typeclasses and verified algebraic instances.

13 Conclusion

We have presented the first machine-checked formalization of braided modular tensor categories, knot invariants, TQFT partition functions, $\mathrm{SU}(3)_k$ fusion categories, the Temperley-Lieb algebra, Jones-Wenzl idempotents, surgery presentations with Kirby moves, and the WRT invariant formula in any proof assistant. The pipeline extends from categorical algebraic data through combinatorial topology to quantum 3-manifold invariants and anyonic quantum gates — with every link machine-checked.

The culminating result is a self-contained universality proof for Fibonacci anyonic braiding: verified MTC data $\to$ braiding matrices over $K = \mathbb{Q}(\zeta_5, \sqrt{\varphi}) \to$ Lie algebra commutators $\to$ density in SU(2) (qubit) and SU(3) (qutrit). The discovery that $\sqrt{\varphi}$ lies outside every cyclotomic field, necessitating a degree-8 non-abelian extension, demonstrates the importance of tracking algebraic number fields precisely — an advantage of formal verification over pen-and-paper calculation.

The key technical enablers — decidable algebraic number fields (including the novel non-cyclotomic K) with `native_decide` verification, and the categorical infrastructure (`PivotalCategory`,

`RibbonCategory`, `FusionCategoryData`, `ModularTensorData`) — are being prepared for contribution to Mathlib.

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